

Customer Behavior Analysis

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PYTHON

SQL

POWER BI



Made with GAMMA

Project Overview

This project analyzes customer purchasing behavior to identify patterns, segment customers, and generate insights that support business decision-making.

Analyze

Identify patterns in buying habits and purchasing behavior

Segment

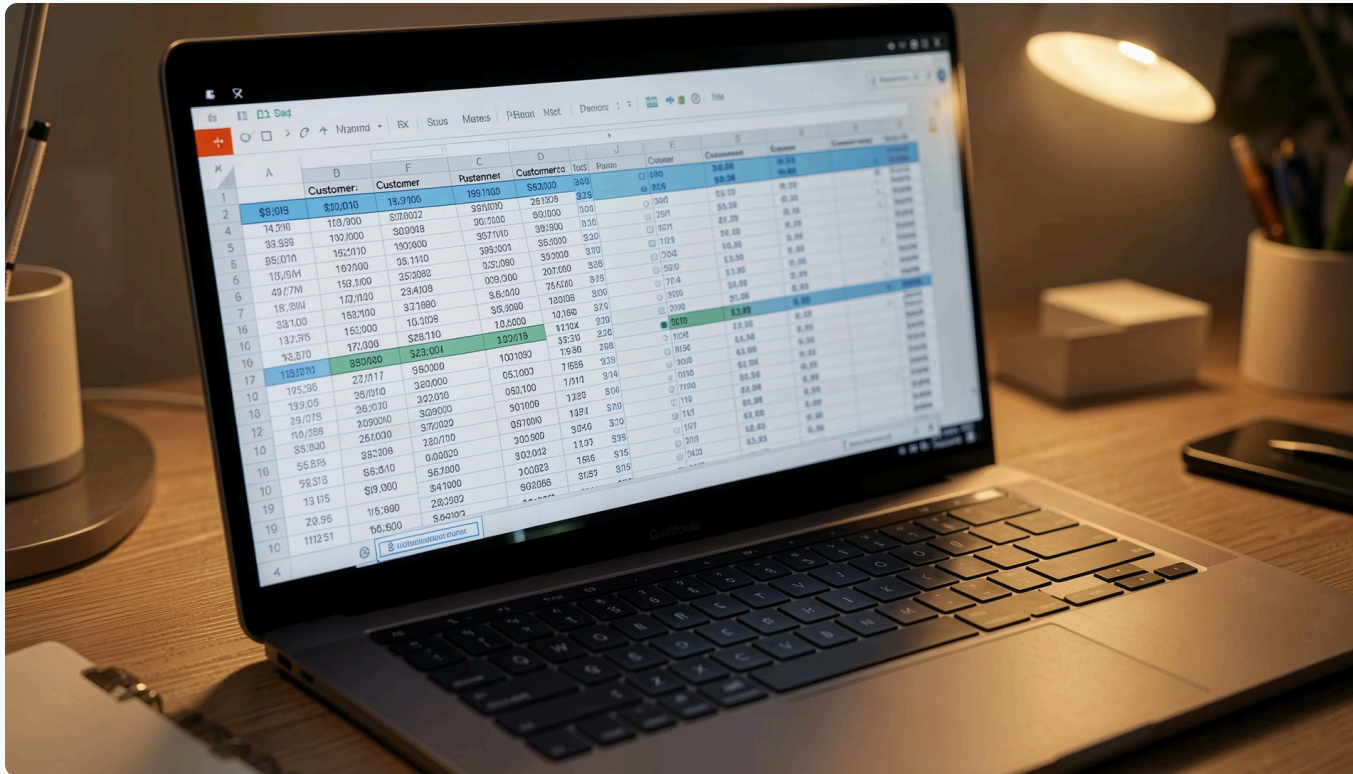
Classify customers by demographics and purchase frequency

Optimize

Support marketing, inventory, and retention strategies



Dataset at a Glance



Customer ID	Age	Gender	Purchase Amount	Category	Review Rating	Subscription Status	Payment Method	Review Date
1	25	Male	\$120.00	Electronics	4.5	Active	Credit Card	2023-10-25
2	30	Female	\$85.00	Clothing	3.8	Active	Debit Card	2023-10-20
3	22	Male	\$150.00	Home Goods	5.0	Active	Credit Card	2023-10-15
4	35	Female	\$95.00	Books	4.2	Active	Debit Card	2023-10-10
5	28	Male	\$110.00	Electronics	4.7	Active	Credit Card	2023-10-05
6	32	Female	\$70.00	Clothing	3.5	Active	Debit Card	2023-09-28
7	27	Male	\$130.00	Home Goods	4.9	Active	Credit Card	2023-09-20
8	31	Female	\$80.00	Books	4.1	Active	Debit Card	2023-09-15
9	24	Male	\$140.00	Electronics	4.6	Active	Credit Card	2023-09-10
10	33	Female	\$90.00	Clothing	3.9	Active	Debit Card	2023-09-05

The dataset contains **3,900 records** across **18 features**, covering customer demographics and transactional information.

Key Variables

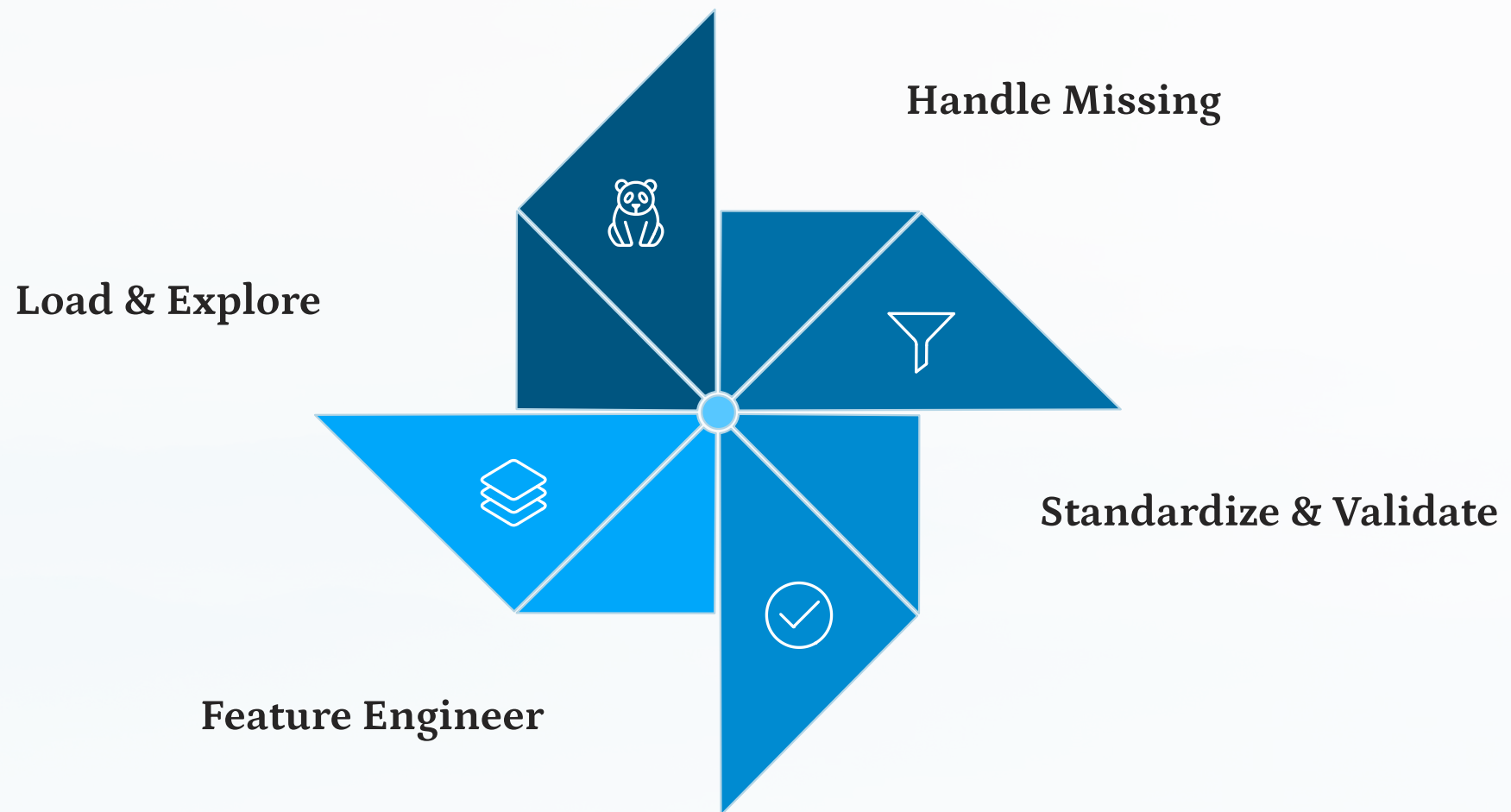
Customer ID, Age, Gender, Purchase Amount, Category, Review Rating, Subscription Status, Payment Method, and more

Data Quality

Mix of numerical and categorical variables. **Review Rating** had 37 missing values, handled during preprocessing.

Data Cleaning & Preparation

A structured preprocessing workflow in Python ensured the dataset was accurate, consistent, and analysis-ready.



Key steps included loading data via Pandas, imputing missing Review Ratings using category-wise medians, standardizing column names, and validating data types. Duplicate detection and consistency checks confirmed no repeated transactions. **Discount Applied** and **Promo Code Used** were found identical, so **Promo Code Used** was removed. The cleaned dataset was then loaded into PostgreSQL via SQLAlchemy for SQL-based analysis.

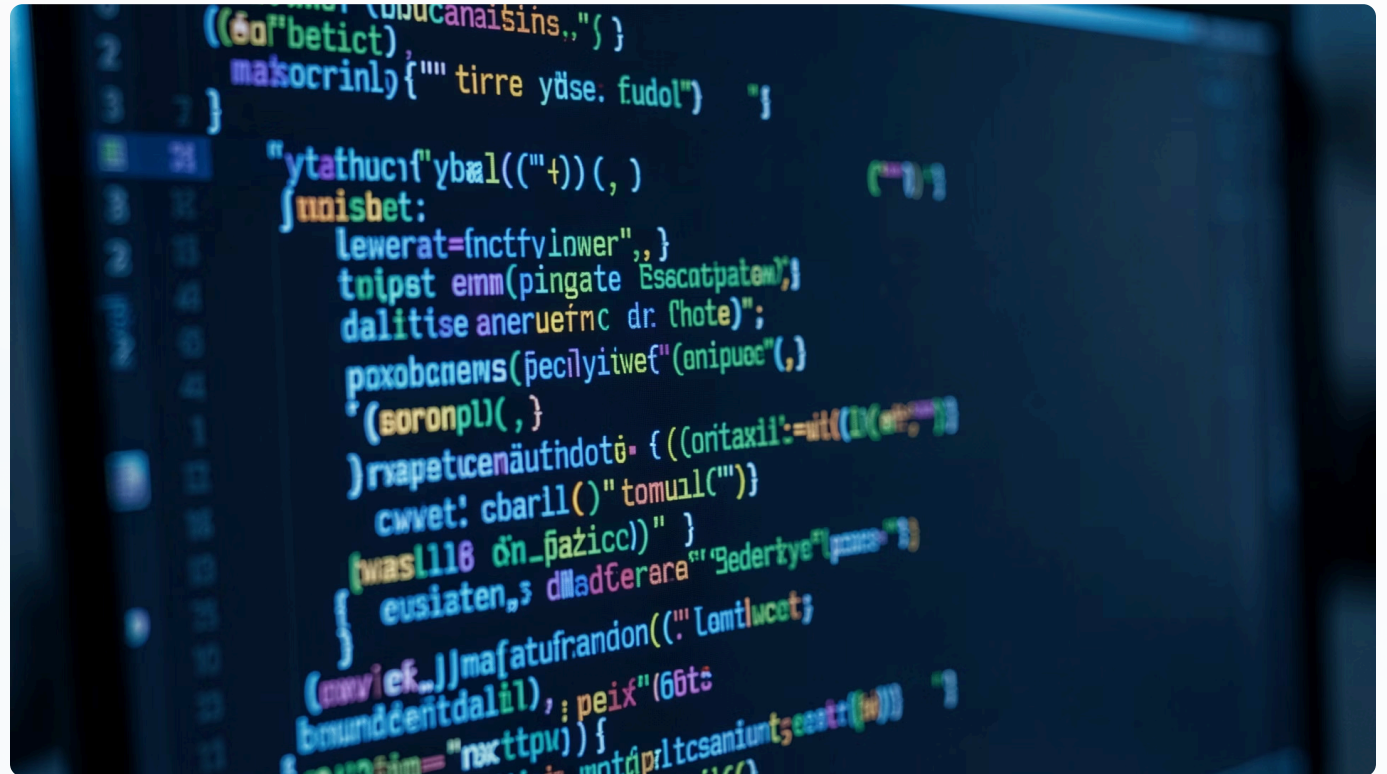
Feature Engineering

Age Group Segmentation

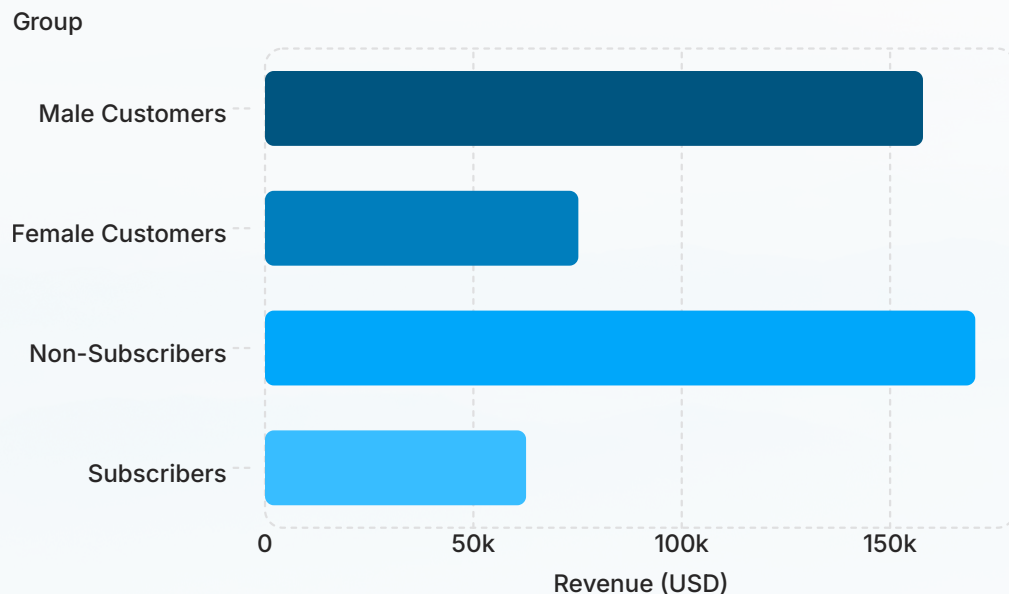
Customers were segmented into **Young Adult, Adult, Middle-aged**, and **Senior** using quantile-based binning for demographic analysis.

Purchase Frequency Transformation

Categorical values (Weekly, Monthly, Fortnightly, etc.) were converted to numerical day equivalents (7, 30, 14...) enabling quantitative behavioral comparison.



SQL Analysis: Revenue Insights



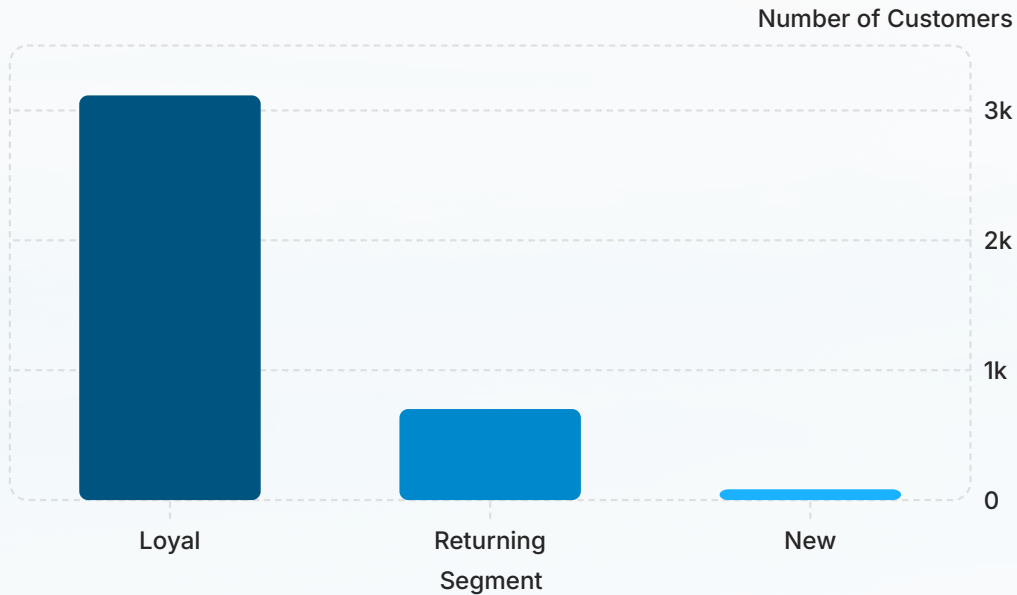
Revenue by Gender

Male customers contributed **\$157,890** vs. **\$75,191** from female customers — more than double the revenue.

Subscriber vs. Non-Subscriber

Non-subscribers (2,847 customers) generated **\$170,436** vs. subscribers (1,053) at **\$62,645**. Average spend was nearly identical (~\$59.50), indicating subscription status had **minimal impact** on individual spending behavior.

Customer Segmentation & Age Group Revenue



Segmentation Results

80% of customers are Loyal (3,116), 701 are Returning, and only 83 are New — strong retention but a clear opportunity to improve new customer acquisition.

Revenue by Age Group

Young Adults led revenue at **\$62,143**, followed by Middle-aged (\$59,197), Adult (\$55,978), and Senior (\$55,763). Revenue distribution is relatively balanced across all groups.

Product Performance



Top 5 Highest-Rated Products

Gloves led with a 3.86 average rating. Small rating differences (3.78–3.86) indicate **consistent customer satisfaction** across top products.

Top Products by Category

- **Accessories:** Jewelry (171 orders), Sunglasses (161), Belt (161)
- **Clothing:** Blouse (171), Pants (171), Shirt (169)
- **Footwear:** Sandals (160), Shoes (150), Sneakers (145)



Power BI Dashboard

An interactive **Customer Behavior Dashboard** was developed to visualize key metrics and support data-driven decisions.

KPIs

3.9K customers · Avg. Purchase: \$59.76 · Avg. Rating: 3.75

Visualizations

Subscription breakdown (27% Yes, 73% No), Sales & Revenue by Category, Revenue & Sales by Age Group

Filters

Subscription Status, Gender, Category, and Shipping Type for dynamic exploration



Business Recommendations

→ **Target High-Value Segments**

Focus marketing on **male and young adult customers** — the highest revenue contributors.

→ **Strengthen Retention**

Offer **loyalty rewards and personalized discounts** to the 3,116 loyal customers.

→ **Acquire New Customers**

Launch **referral programs** and **first-time discounts** to grow the small new-customer base (83).

→ Optimize Inventory & Subscriptions

Stock up on **high-performing categories** (Clothing, Footwear) and enhance subscription benefits to drive sign-ups.